

Rakesh Kumar

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EXPERIENCE

AI Backend Engineer, Lazywork

2026 – Present

GroundedDocs — private document Q&A that answers only with faithful citations from the user's files, and abstains when evidence is insufficient.

- Designed idempotent PDF ingest with content hashing and stable chunk IDs so a changed document re-indexes without duplicate vectors or stale citations; treated parse and chunk boundaries as product decisions, not library defaults.
- Built authorization-aware hybrid retrieval (pgvector + BM25 + RRF, then cross-encoder rerank) scoped by `user_id` and workspace from day one, so unscoped search is a defect.
- Shipped structured answer-with-citations and calibrated abstention as the default path, held on a frozen 40-question set (18 answerable / 11 partial / 11 unanswerable).
- Modeled threats with DFD + STRIDE; fail-closed auth, rate limits, and secret isolation. Treated retrieved document text as untrusted input and ran a separate ≥ 30 -case adversarial suite covering planted instructions, prompt extraction, cross-user access, hostile PDFs, and conflicting sources.
- Added a bounded LangGraph path only for question types hybrid RAG still failed; loop limits, one read-only MCP document-lookup tool, at most one ephemeral subagent — kept only when it beat deterministic RAG on quality, latency, cost, and failure rate.
- Built eval gates separating retrieval quality, citation faithfulness, abstention precision/recall, security pass rate, structured-output success, latency, and cost per successful task, with version tags so prompt or corpus changes cannot drift silently.
- Put an OpenAI-compatible gateway in front of the app across two or more providers: normalized streaming and structured outputs, tenant API-key auth, token budgets, timeouts, bounded retries, circuit breakers, and privacy-aware cache rules. Measured cost per successful task after failure injection.

Backend Java Developer, Genpact (Client: Shutterfly, USA)

Oct 2022 – Present

- Architected Java 11 / Spring Boot microservices processing 2 million daily transactions, reducing API response times by 65%.
- Implemented Resilience4j across 15+ microservices and reached 99.9% system uptime; optimized PostgreSQL, MongoDB, and Redis, increasing retrieval speed by 70% and cutting costs by 30%.
- Built React.js + TypeScript internal tools and a photo pipeline for 5 million daily uploads with sub-200 ms responses and compression that cut storage cost by 40%.
- Automated release management in TypeScript (85% less manual coordination) and mentored 3 developers, raising code coverage from 65% to 92% through TDD.
- Backend / Full Stack Engineer**, Digital Asset Management Client engagement Engineered a high-performance photo processing pipeline managing over 5 million daily uploads with sub-200 ms response times. Developed a React.js frontend with TypeScript for image tools, implementing smart compression that reduced storage costs by 40%.
- Full Stack Developer**, Internal Tools & Automation Platform Created a React.js application with pagination APIs capable of handling 100k+ records, improving search performance by 60%. Automated release management with a TypeScript service, reducing manual coordination effort by 85%.

TECHNICAL SKILLS

Languages	Python, Java 8/11/17, TypeScript, JavaScript (ES6+), SQL
AI / Backend	FastAPI, Spring Boot, hybrid retrieval (pgvector + BM25 + RRF), LangGraph, MCP, structured outputs
Data	PostgreSQL, pgvector, MongoDB, Redis, SQL optimization
DevOps	Docker, Kubernetes, AWS (EC2, RDS, S3), Jenkins, CI/CD, Git
Quality	Custom eval harness, JUnit 5, Mockito, TDD, Resilience4j

EDUCATION

National Institute of Technology (NIT) Bhopal

Jan 2018 – Jan 2022

Bachelor of Technology (B.Tech)

ADDITIONAL

- Leadership:** Led ISTE chapter initiatives at NIT Bhopal, organizing technical events for 200+ students; won 2 internal hackathons.
- Community:** Volunteered 100+ hours at Arushi NGO, Bhopal, working with specially abled children.